

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460325.9248 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.171 +/- 0.01
Transit depth (Rp/Rs)^2	2.93 +/- 0.36 [%]
Orbital Inclination (inc)	82.2 +/- 0.99
Airmass coefficient 1 (a1)	1.0506 +/- 0.0081
Airmass coefficient 2 (a2)	-0.034 +/- 0.036
Scatter in the residuals of the lightcurve fit is	0.77 %
Best Comparison Star	#2 - [589, 290]
Optimal Aperture	4.4
Optimal Annulus	13.66
Transit Duration (day)	0.0519 +/- 0.0031

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.171 ± 0.01 vs 0.1594 (deviation 1.16σ)
Duration fit vs published	0.0519 d vs 0.0512 d (deviation 0.21σ)
Mid-time O–C	-0.7 min
Depth SNR	17.1
Residual scatter	0.77 %

Light curve

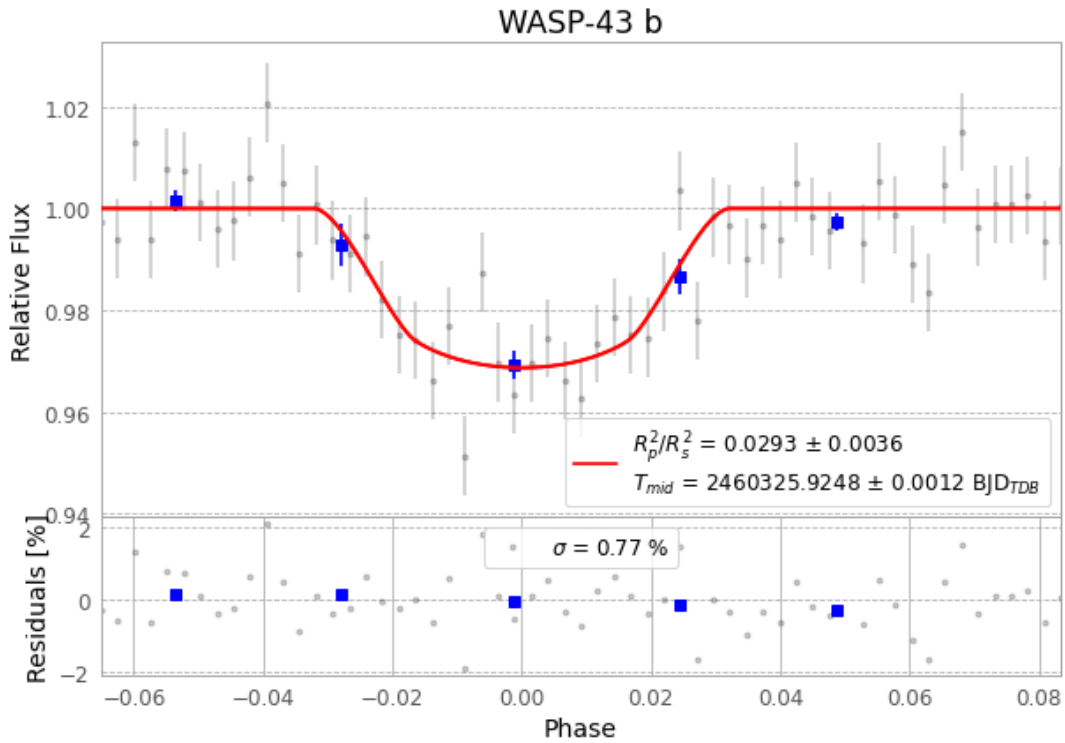


Figure 1 — FinalLightCurve_WASP-43 b_2024-01-16.png (SHA-256 efe3108f45ce0292...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [501, 244], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c536dac13becb7b7...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

59 FITS frames (39 MB), WASP-43240116084815.FITS → WASP-43240116114215.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-240116.log (86cfe296689b...), FinalLightCurve_WASP-43 b_2024-01-16.png (efe3108f45ce...), FinalLightCurve_WASP-43 b_2024-01-16.csv (a499cb172c17...)

Compute resources

Wall time 150.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-20 00:05Z.